

What Does Adaptivity Cost? Measuring the Exploitation–Exploitability Frontier of Opponent-Conditioned Policies in 3-Player Leduc Hold'em

Undergraduate Research Plan

1. Research Question

Shi, Guo, Liu and Fan (2025) propose AMP3, an adaptive multi-player poker policy learned via Actor-Critic reinforcement learning conditioned on predicted opponent style features. Their introduction draws a distinction that motivates the whole method: an *optimal* policy is unexploitable but cannot maximize return against any specific opponent, while a *best* policy maximally exploits a specific opponent but is itself exploitable and fails to generalize.

They assert this tradeoff. They do not measure it, because six-player No-Limit Hold'em is far too large for exploitability to be computed.

Our question: In a game small enough to compute both quantities exactly, what is the shape of the tradeoff between exploitation gain and individual best-response exploitability for an opponent-style-conditioned policy?

Supporting questions:

- **(B)** Does a style-conditioned policy generalize to opponent styles held out of training, or has it learned a lookup table over the training style library?
- **(C)** How many observed hands does opponent style modeling require before its predictions are useful to the policy?
- **(A, if time allows)** Is the performance gain attributable to style *information*, or merely to the added network capacity of the conditioning pathway?

2. Positioning and Honest Scoping

Two limitations must be stated explicitly in the paper, not buried.

This is reimplementing from description, not replication. No code release accompanies Shi et al. (2025). Unless the authors respond to a code request, we implement AMP3 from the published description. We therefore do not claim to reproduce their numbers, and we do not claim their method fails if ours underperforms. We test *the idea*, not *their artifact*.

The style features are redesigned, not ported. Standard Hold'em style statistics (VIP, PFR, 3-bet frequency, WTSD) have no clean analogue in Leduc, which has two betting rounds, one private card and one board card. We define an analogous feature set for Leduc (Section 4.2). This is a deliberate adaptation and a small contribution in itself, but it means our agent is not their agent.

Both caveats are acceptable for a workshop paper provided they appear in the abstract and the limitations section.

3. Formal Setup

3.1 Individual Best-Response Exploitability (No Collusion)

For a strategy profile $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ in 3-player Leduc, define the individual exploitability of player i :

$$\text{Exploit}_i(\sigma) = \max_{\sigma'_i} u_i(\sigma'_i, \sigma_{-i}) - u_i(\sigma)$$

and total exploitability as the sum over the three players. The two non-deviating players hold their strategies fixed and do not coordinate. This is the no-collusion assumption: we measure what a single opponent could gain by deviating unilaterally, which is both the computable quantity and the one that matters under real poker rules.

3.2 Exploitation Gain

Let Π_{style} be a pool of scripted style opponents. The exploitation gain of policy π is its expected return against that pool, measured in milli-big-blinds per hand, relative to an equilibrium baseline:

$$G(\pi) = \mathbb{E}_{\text{opp} \sim \Pi_{\text{style}}} [u(\pi, \text{opp})] - \mathbb{E}_{\text{opp} \sim \Pi_{\text{style}}} [u(\pi_{\text{CFR}}, \text{opp})]$$

where π_{CFR} is a tabular CFR approximate equilibrium.

3.3 The Frontier

We need a knob that varies adaptation strength continuously. Primary mechanism:

$$\pi_{\lambda} = (1 - \lambda) \pi_{\text{CFR}} + \lambda \pi_{\text{AMP3}}, \quad \lambda \in [0, 1]$$

Sweeping λ and plotting $G(\pi_{\lambda})$ against $\text{Exploit}(\pi_{\lambda})$ traces the frontier. At $\lambda = 0$ we recover the equilibrium baseline (high exploitability floor near zero, no exploitation gain); at $\lambda = 1$ we get the fully adaptive policy.

Secondary mechanism (if time allows): vary the entropy regularization coefficient in the Actor-Critic objective, which produces a frontier through training rather than through post-hoc mixing. Comparing the two frontiers is itself interesting: a training-time frontier that dominates the mixing frontier would be a genuine finding.

3.4 A Technical Subtlety: Exploitability of an Adaptive Agent

An agent that updates its style estimate online is a **non-stationary** strategy, and computing a best response to it is a partially observable planning problem, not a tree traversal. Standard exploitability is undefined.

Our protocol: freeze the style input. For each style s in the library, condition the policy on s , freeze it into a stationary strategy $\pi(\cdot \mid s)$, and compute exploitability exactly. Report:

- Exploitability as a function of s
- Worst-case exploitability, $\max_s \text{Exploit}(\pi(\cdot \mid s))$
- Mean exploitability over the style library

This is a scoping decision that must be stated. It measures the exploitability of the policies the agent *commits to*, not of the adaptive procedure as a whole. That is the right question here and it is the only tractable one.

4. Environment and Opponent Design

4.1 Game

- **3-player Leduc Hold'em** via DeepMind `open_spiel`: `pyspiel.load_game("leduc_poker", {"players": 3})`
- 6-card deck (J, Q, K x2), one private card each, one board card, two betting rounds, fixed bet sizes, one raise per round
- Approximately 10^3 information sets, so exact best-response computation runs in seconds

4.2 Leduc Style Features

Replacing the Hold'em HUD statistics with Leduc analogues. Each is a scalar in $[0, 1]$ computed from an opponent's observed action history:

Feature	Definition	Hold'em analogue
<code>vpip_r1</code>	Frequency of calling or raising in round 1	VPIP
<code>agg_r1</code>	raises / (raises + calls) in round 1	PFR / aggression factor
<code>agg_r2</code>	raises / (raises + calls) in round 2	Postflop aggression
<code>fold_to_raise</code>	Frequency of folding when facing a raise	Fold-to-cbet
<code>showdown_rate</code>	Frequency of reaching showdown given round 1 entry	WTSD
<code>bluff_rate</code>	Raise frequency holding an unpaired J	Bluff frequency

The first two dimensions map onto the loose/tight and aggressive/passive axes used in the original paper's style taxonomy.

4.3 Style Library

Twelve scripted opponents. Each is a parameterized rule-based agent defined by thresholds over (private card, board card, betting history).

Training styles (8): the 2x2 grid of {loose, tight} x {aggressive, passive}, each at two intensity levels.

Held-out styles (4), used only in Experiment B:

- *Calling station*: never folds, never raises
- *Maniac*: raises at maximum frequency regardless of hand
- *Trap player*: passive with strong hands, aggressive with weak (inverted aggression)
- *CFR-equilibrium player*: the tabular CFR solution itself, which belongs to no style category

The trap player and the CFR player are the interesting held-out cases. The first is adversarial to the style representation; the second is out-of-distribution in a different way, being not a “style” at all.

4.4 Style Encoding: A Design Choice to Test

Two options, and we test both because the choice is not obvious:

1. **Discrete classification.** OSM predicts a categorical style label; the policy is conditioned on a one-hot vector. Simpler, but cannot represent unseen styles, which likely hurts on Experiment B.
2. **Continuous feature regression.** OSM predicts the six-dimensional feature vector directly; the policy is conditioned on that vector. More expressive and should generalize better.

Hypothesis: the continuous encoding generalizes to held-out styles and the discrete one does not. If confirmed, this is a concrete, actionable design recommendation, which is exactly what a workshop audience wants from a small-scale empirical paper.

5. Experiments

Experiment E (headline): The Frontier

Procedure. Train AMP3 against the 8 training styles. Sweep λ over $\{0, 0.1, 0.2, \dots, 1.0\}$. At each λ : measure exploitation gain against the training pool, and measure exploitability under the freeze-and-condition protocol of Section 3.4.

Output. A scatter/line plot with exploitability on the x-axis and exploitation gain on the y-axis, one point per λ , error bars over 5 seeds. This is Figure 1 of the paper.

What we learn. Is the tradeoff linear, convex, or does a knee exist? A convex frontier with an early knee would mean most of the exploitation gain is available at low exploitability cost, which is a genuinely useful practical finding.

Experiment B: Held-Out Style Generalization

Procedure. Evaluate the trained policy against the 4 held-out styles. Compare against three baselines: tabular CFR (unadaptive), an oracle policy trained directly on the held-out style, and the policy conditioned on the *nearest training style* rather than its own OSM prediction.

Output. Table of mbb/hand against each held-out opponent, discrete vs. continuous encoding.

What we learn. The gap between the AMP3 policy and the oracle quantifies the generalization failure. The comparison against nearest-training-style tells us whether OSM’s predictions carry information beyond crude nearest-neighbor matching.

Experiment C: Observation Budget

Procedure. Vary the number of observed opponent hands available to OSM: {5, 10, 25, 50, 100, 250, 500}. Measure two things at each budget: style prediction accuracy (or feature MSE for the continuous encoding), and downstream exploitation gain.

Output. Two learning curves on shared axes.

What we learn. The break-even point, meaning the number of hands after which adaptation beats playing the equilibrium. This is the most practically relevant number in the paper and is likely small in Leduc, which is worth stating as a limitation for transfer to full Hold’em.

Experiment A (if time allows): Permutation Control

Procedure. Retrain with style labels randomly permuted across opponents. Identical architecture, identical parameter count, zero style information.

Output. One row added to the main results table.

What we learn. Whether the gain is from style information or capacity. A null result here would substantially weaken the case for OSM and is worth knowing.

6. Timeline

Assumes 2–3 people at roughly 5 hours per week each, so approximately 100–150 person-hours total.

Week	Work	Deliverable
1	Obtain both papers. Read Shi et al. (2022) <i>Entropy</i> paper in full for the Actor-Critic base architecture. Install <code>open_spiel</code> , load 3-player Leduc, run tabular CFR to convergence, verify exploitability approaches zero.	Working CFR baseline with exploitability logging
2	Read Shi et al. (2025) for the OSM component. Implement the 12 scripted style opponents and the 6 Leduc style features. Verify each opponent’s features match its intended archetype.	Style library + feature extractor, with a validation table
3	Implement the OSM network (small MLP over aggregated action-history features). Train and evaluate on style classification / feature regression against the 8 training opponents.	OSM network hitting reasonable accuracy on training styles
4	Implement the Actor-Critic policy: Actor with imperfect information, Critic with perfect information, following the 2022 paper. Train without style conditioning first as a sanity check.	Unconditioned RL agent that beats random and loses to CFR
5	Add style conditioning. Train AMP3 proper, both discrete and continuous encodings. Debug convergence. Expect this week to overrun.	Trained AMP3 agents, 5 seeds each
6	Experiment E. Implement the freeze-and-condition exploitability protocol. Sweep λ . Produce the frontier plot.	Figure 1, the headline result

Week	Work	Deliverable
7	Experiments B and C. Both reuse the week 6 infrastructure and should be fast.	Held-out table, observation-budget curves
8	Buffer week. Run Experiment A if on schedule; otherwise absorb slippage from week 5. Additional seeds for anything with wide error bars.	Complete results
9	Write the paper. Draft in order: methods, results, then introduction and abstract last.	Full draft
10	Revise, tighten, format for the target venue, prepare code release.	Submission

Critical path: weeks 4–5. If the Actor-Critic implementation does not converge by end of week 5, cut Experiment A and the entropy-regularization frontier immediately, and consider replacing the learned policy with a simpler style-conditioned policy (for instance, tabular best-responses to each style, mixed by OSM’s predicted style distribution). That fallback still supports Experiments E, B and C, and preserves the headline result. **Decide this at the week 5 checkpoint, not later.**

7. Risks and Mitigations

Risk	Likelihood	Mitigation
Cannot obtain Shi et al. (2025) full text	Low	University library; the 2022 <i>Entropy</i> paper is open access and carries most of the architecture
Authors do not share code	High	Plan assumes reimplementation from the start; state it in the paper

Risk	Likelihood	Mitigation
Actor-Critic fails to converge in 3-player Leduc	Medium	Week 5 checkpoint with the tabular best-response fallback described above
Frontier is trivially linear, no interesting structure	Medium	Still a publishable measurement; it confirms or refutes a claim nobody has tested. Report it either way
Leduc is too small for style differences to matter	Medium	Validate in week 2 that the 12 opponents differ meaningfully in win rate against CFR. If they do not, enlarge the game (3-player limit Hold'em) or add betting rounds
Results are all negative	Low concern	Negative results with clean methodology are workshop-appropriate, especially when testing an untested claim from a published paper

The fifth row is the one to watch. Run that validation in week 2, before building anything on top of the style library.

8. Engineering Stack

Component	Choice
Game environment	<code>open_spiel</code> (DeepMind, Apache 2.0), 3-player Leduc
Equilibrium baseline	<code>open_spiel</code> tabular CFR / CFR+
Exploitability	<code>open_spiel</code> best-response utilities
Neural networks	PyTorch; small MLPs only (2–3 layers, 64–128 units)
Style opponents	Custom Python, rule-based, in-repo
Experiment tracking	Weights & Biases or CSV plus a config hash
Analysis	pandas, matplotlib

Component	Choice
Compute	CPU sufficient throughout. No GPU required

Keep the architecture deliberately small. The research question is about the exploitation and exploitability tradeoff, not about architecture search, and a large network would only add confounds.

9. Paper Outline

Target length 4–6 pages plus references.

1. **Introduction.** The optimal-versus-best policy tradeoff as posed by Shi et al. State that it has been asserted but not measured, and that measuring it requires scaling down.
2. **Related work.** CFR and exploitability; opponent modeling in poker; the AMP3 / OSM line of work; prior opponent-modeling-plus-CFR hybrids.
3. **Setup.** 3-player Leduc; individual best-response exploitability under no collusion; the freeze-and-condition protocol; Leduc style features and the style library.
4. **Method.** Reimplementation of AMP3; discrete versus continuous style encoding; the λ -mixing frontier.
5. **Results.** Figure 1 (frontier), Table 1 (held-out styles), Figure 2 (observation budget), Table 2 (permutation control, if run).
6. **Discussion.** Implications for practical multiplayer solver design; what transfers to larger games and what does not.
7. **Limitations.** Reimplementation not replication; redesigned features; Leduc’s small size; the freeze-and-condition scoping of exploitability.

Target venues: NeurIPS workshops on decision-making or on RL in games; AAAI Workshop on Reinforcement Learning in Games; ICML workshops in the same space. AAAI’s student abstract track is a reasonable secondary target.

10. Reading List

Priority	Reference	Purpose
Essential	Shi, Guo, Liu & Fan (2022), <i>Optimal Policy of Multiplayer Poker via Actor-Critic Reinforcement Learning</i> , Entropy	Open access; the Actor-Critic base architecture, APU and Dual-APU
Essential	Shi, Guo, Liu & Fan (2025), <i>Adaptive multi-player poker policy learning based on opponent style modeling</i> , Neural Computing and Applications 37(19)	The paper being extended; OSM and AMP3
Essential	Zinkevich et al. (2007), <i>Regret Minimization in Games with Incomplete Information</i>	CFR and the exploitability baseline
Important	Brown & Sandholm (2019), <i>Superhuman AI for Multiplayer Poker</i> , Science	Multiplayer context; why equilibrium is problematic beyond two players
Important	Johanson et al. (2011), <i>Accelerating Best Response Calculation</i>	Best-response computation, relevant to the measurement protocol
Useful	Yuan et al. (2021), <i>Opponent-Restricted Response Solving on Texas Hold'em Poker</i>	Prior work explicitly trading off exploitability against exploitation; closest existing framing to ours
Useful	Ganzfried & Sandholm, work on opponent modeling with safety guarantees	Theoretical treatment of bounded-exploitability exploitation
Optional	Brown et al. (2019), <i>Deep Counterfactual Regret Minimization</i>	Context on neural approximation in this space

Yuan et al. (2021) deserves particular attention during week 1. It addresses the same tradeoff from the CFR side rather than the RL side, and needs to be distinguished from our contribution in the related work section.

11. First Three Actions

1. Request both papers through the university library today. Email the authors for code at the same time, since a reply may take weeks.
2. Install `open_spiel`, load 3-player Leduc, run tabular CFR, and confirm exploitability approaches zero. This is the sanity check that the whole project rests on and takes an afternoon.
3. Sketch the 12 style opponents on paper before writing any code, and predict which should beat which. If the predictions are not obviously differentiated, the style library needs redesign before week 3.